

Understanding Common Synthesis Error in Yosys

INTRODUCTION:

Synthesis is the process of converting RTL design written in Verilog into a gate-level netlist using standard-cell libraries. During this process, errors often arise that prevent successful synthesis. Yosys, an open-source synthesis tool, reports such issues through error messages that must be analysed and resolved. One of the key components in synthesis is the Liberty (.lib) file, which defines the functionality, timing, and power characteristics of standard cells. Errors in the RTL code or in specifying the library path are among the most common problems encountered. By studying these errors, the causes can be identified and corrective measures can be applied, ensuring a smooth synthesis flow.

OBJECTIVE:

After completing this lab, you will be able to:

- Recognition of common synthesis errors that occur in Yosys.
- Understanding of RTL syntax-related issues and their corrections.
- Identification of library path errors during synthesis.
- Learning corrective methods to resolve these errors.

COMMON ERRORS :

Error 1: RTL Syntax Error

- After Invoking Yosys , we load the verilog file by using read_verilog command

```
Yosys 0.9 (git sha1 1979e0b)

-- Executing script file `script.js' --

1. Executing Verilog-2005 frontend: design.v
Parsing Verilog input from `design.v' to AST representation.
design.v:7: ERROR: syntax error, unexpected TOK_ASSIGN, expecting ';' ;'
```

- Sometimes due to error in the verilog, this error encounters

```

1 module decoder_expr (
2     input en,
3     input [1:0] a,
4     output [3:0] y
5 )
6
7 assign y[0] = en & ~a[1] & ~a[0];
8 assign y[1] = en & ~a[1] & a[0];
9 assign y[2] = en & a[1] & ~a[0];
10 assign y[3] = en & a[1] & a[0];
11
12 endmodule
    
```

- To fix this error, we need to debug the RTL and correct the syntax mistake. In this example, the issue is in the 5th line where a semicolon is missing. The solution is to add a semicolon ; after the closing parenthesis) on the 5th line.

```

1 module decoder_expr (
2     input en,
3     input [1:0] a,
4     output [3:0] y
5 );|
6
7 assign y[0] = en & ~a[1] & ~a[0];
8 assign y[1] = en & ~a[1] & a[0];
9 assign y[2] = en & a[1] & ~a[0];
10 assign y[3] = en & a[1] & a[0];
11
12 endmodule
13
    
```

Error 2 : Library Path Error

- Sometimes this library mapping error occurs due to an incorrect library path specified in the command.

```

7. Executing DFFLIBMAP pass (mapping DFF cells to sequential cells from liberty file).
ERROR: Can't open liberty file `0su018_stdcells.lib': No such file or directory
    
```

- To resolve this, we need to verify the library path / correct library name.

```
dfflibmap -liberty osu018_stdcells.lib
```

```
abc -liberty osu018_stdcells.lib
```

OBSERVATION :

- read_verilog Syntax Error:** When attempting to read the Verilog file, Yosys reported a syntax error. This occurred due to incorrect Verilog syntax in the design file, such as missing semicolons, mismatched parentheses, or unsupported constructs.

- **Library Path Error:** During synthesis, Yosys could not locate the standard cell library because the specified library path was incorrect or missing. This caused synthesis to fail as the tool was unable to map RTL to standard cells.

CONCLUSION :

This experiment demonstrated common synthesis errors in Yosys and their causes. The `read_verilog` syntax error highlighted the importance of writing correct and tool-compatible Verilog code. The library path error emphasized the need to correctly specify library locations for proper cell mapping. Understanding these errors helps in debugging RTL designs and ensures smooth synthesis workflow.